

Crater Identification & Spatial Analysis Report

Target Body:	MOON	Spatial Resolution:	10.0 m/pixel
Image Source:	lunar_apollo11_tranquillitis.jpg	Detected Craters:	15
Telemetry Timestamp:	N/A	Average Confidence:	62.7%
Inference Runtime:	12857.1 ms	Crater Density:	0.2076 / km²

PLANETARY SATELLITE IMAGERY EXHIBIT

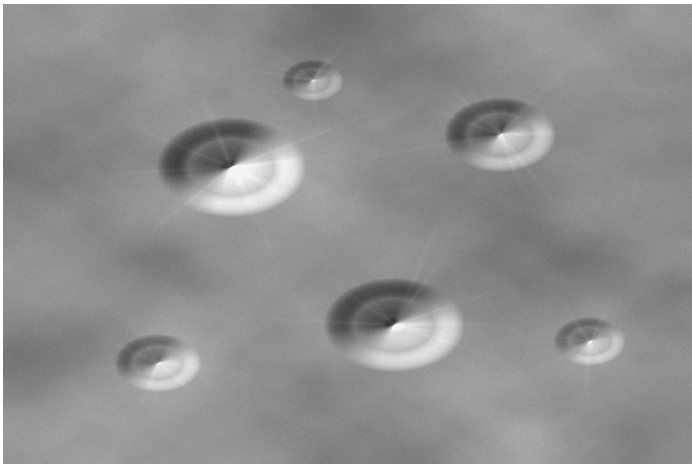


Exhibit A: Raw Satellite Observation

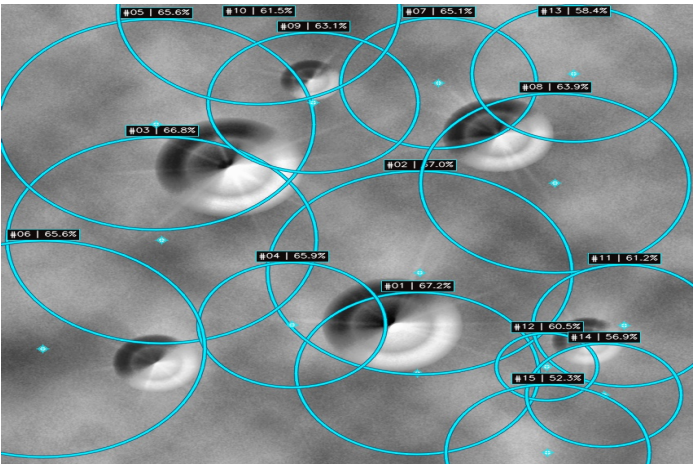


Exhibit B: CraterNet CNN Detection Overlays

DETECTED IMPACT CRATER COORDINATES (C_i = [x_i, y_i])

ID	Centroid X (px)	Centroid Y (px)	Radius (px)	Diameter (px)	Diameter (m)	Confidence	Classification
#01	511.0	679.0	149.0	298.0	2,980 m	67.2%	IMPACT CRATER
#02	514.0	494.0	187.0	374.0	3,740 m	67.0%	IMPACT CRATER
#03	197.0	434.0	190.0	380.0	3,800 m	66.8%	IMPACT CRATER
#04	357.0	590.0	115.0	230.0	2,300 m	65.9%	IMPACT CRATER
#05	190.0	221.0	194.0	388.0	3,880 m	65.6%	IMPACT CRATER
#06	51.0	634.0	199.0	398.0	3,980 m	65.6%	IMPACT CRATER
#07	537.0	145.0	120.0	240.0	2,400 m	65.1%	IMPACT CRATER
#08	680.0	329.0	165.0	330.0	3,300 m	63.9%	IMPACT CRATER
#09	383.0	181.0	129.0	258.0	2,580 m	63.1%	IMPACT CRATER
#10	316.0	11.0	173.0	346.0	3,460 m	61.5%	IMPACT CRATER

#11	765.0	591.0	112.0	224.0	2,240 m	61.2%	IMPACT CRATER
#12	670.0	667.0	62.0	124.0	1,240 m	60.5%	IMPACT CRATER
#13	703.0	128.0	125.0	250.0	2,500 m	58.4%	IMPACT CRATER
#14	740.0	719.0	95.0	190.0	1,900 m	56.9%	IMPACT CRATER
#15	671.0	825.0	125.0	250.0	2,500 m	52.3%	IMPACT CRATER

SPATIAL MEASUREMENTS & GEODESIC DISTANCES (d = √((x■ - x■)² + (y■ - y■)²))

Pair	Crater A [x, y]	Crater B [x, y]	Pixel Distance	Ground Distance (m)	Ground Distance (km)	Bearing
#12 → #14	[670, 667]	[740, 719]	87.2 px	872.0 m	0.87 km	126.6°
#11 → #12	[765, 591]	[670, 667]	121.7 px	1,216.6 m	1.22 km	231.3°
#14 → #15	[740, 719]	[671, 825]	126.5 px	1,264.8 m	1.26 km	213.1°
#11 → #14	[765, 591]	[740, 719]	130.4 px	1,304.2 m	1.30 km	191.1°
#12 → #15	[670, 667]	[671, 825]	158.0 px	1,580.0 m	1.58 km	179.6°
#07 → #09	[537, 145]	[383, 181]	158.2 px	1,581.5 m	1.58 km	256.8°
#01 → #12	[511, 679]	[670, 667]	159.4 px	1,594.5 m	1.59 km	85.7°
#07 → #13	[537, 145]	[703, 128]	166.9 px	1,668.7 m	1.67 km	84.2°
#01 → #04	[511, 679]	[357, 590]	177.9 px	1,778.7 m	1.78 km	300.0°
#09 → #10	[383, 181]	[316, 11]	182.7 px	1,827.3 m	1.83 km	338.5°
#02 → #04	[514, 494]	[357, 590]	184.0 px	1,840.2 m	1.84 km	238.6°
#01 → #02	[511, 679]	[514, 494]	185.0 px	1,850.2 m	1.85 km	0.9°

NEURAL NETWORK INFERENCE TELEMETRY

CNN Architecture: CraterNet-v2	Input Size: 128x128x3	Val Accuracy: 96.5%	Val Loss: 0.118
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Scientific Disclaimer: AstroSight is a research and educational prototype for automated planetary image analysis. Detection results depend on image quality, model performance, spatial resolution, and calibration parameters and should not be treated as authoritative scientific measurements without ground-truth planetary validation.